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Lecture 30 4/6/26

Summary of Last Week:

- Finished Calculus of variations $\Rightarrow$ $\frac{\partial f}{\partial y_i} = \frac{d}{dx}\frac{\partial f}{\partial y_i'}$ $\forall$ variables y that can be varied over
- Started Ch. 7: Lagrangians!

$S = \int \mathcal{L}\, dt$ $\Rightarrow$ extrema of S yield equations of motion

$\mathcal{L} \equiv T - U$ & EOM's come from $\mathcal{L}$-equations $\frac{\partial \mathcal{L}}{\partial q_i} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_i}$

Found that in Cartesian coordinates, for a free particle:

$$\frac{\partial \mathcal{L}}{\partial x_i} = -\frac{\partial U}{\partial x_i} = +F_i \qquad \frac{\partial \mathcal{L}}{\partial \dot{x}_i} = \frac{\partial}{\partial \dot{x}_i}(T) = \frac{\partial}{\partial \dot{x}_i}\left(\tfrac{1}{2}m\,\dot{\vec{x}}\cdot\dot{\vec{x}}\right) = m\dot{x}_i$$

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}_i} = m\ddot{x}_i$$

$\Rightarrow$ $F_i = m\ddot{x}_i$ $\Rightarrow$ Newton's 2nd Law

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Hamilton's Principle: The path a particle follows between points 1 & 2 in the time interval t_1, t_2 is such that the action is stationary:

$$S = \int_{t_1}^{t_2} \mathcal{L}(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n)\, dt$$

The action

Hamilton's principle lets us use $\mathcal{L}$ equations hold in any coordinate system.

We can use Cartesian (x, y, z); Cylindrical (ρ, ϕ, z); Spherical (r, θ, ϕ), etc.

Consider a generalized coordinate system q_1, q_2, q_3:

$q_i = q_i(\vec{r})$, $i = 1, 2, 3$

$\vec{r} = \vec{r}(q_1, q_2, q_3)$

[Each $\vec{r}$ gives a unique q_1, q_2, q_3 and each (q_1, q_2, q_3) gives a unique $\vec{r}$]

$$\mathcal{L} = \tfrac{1}{2}m\,\dot{\vec{r}}^{\,2} - U(\vec{r}) = \mathcal{L}(q_1, q_2, q_3, \dot{q}_1, \dot{q}_2, \dot{q}_3)$$

$$S = \int_{t_1}^{t_2} \mathcal{L}(q_1, q_2, q_3, \dot{q}_1, \dot{q}_2, \dot{q}_3)\, dt$$

S = stationary $\Rightarrow$

$$\frac{\partial \mathcal{L}}{\partial q_1} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_1}$$

$$\frac{\partial \mathcal{L}}{\partial q_2} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_2}$$

$$\frac{\partial \mathcal{L}}{\partial q_3} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_3}$$

$$\Rightarrow \quad \underbrace{\frac{\partial \mathcal{L}}{\partial q_i}}_{\text{generalized force}} = \frac{d}{dt}\underbrace{\frac{\partial \mathcal{L}}{\partial \dot{q}_i}}_{\text{generalized momentum}}$$

Example: Polar coordinates:

$$\mathcal{L} = T - U = \tfrac{1}{2} m \vec{v}\cdot\vec{v} - U(r,\phi) \qquad \vec{v} = \dot{r}\hat{r} + r\dot{\phi}\hat{\phi}$$

$$= \tfrac{1}{2} m (\dot{r}^2 - r^2\dot{\phi}^2) - U(r,\phi)$$

Motion is determined by: $\dfrac{\partial \mathcal{L}}{\partial r} = \dfrac{d}{dt}\dfrac{\partial \mathcal{L}}{\partial \dot{r}}$ & $\dfrac{\partial \mathcal{L}}{\partial \phi} = \dfrac{d}{dt}\dfrac{\partial \mathcal{L}}{\partial \dot{\phi}}$

<u>r-equation</u>:

$$\frac{\partial \mathcal{L}}{\partial r} = \frac{\partial}{\partial r}\left[\tfrac{1}{2} m \dot{r}^2 + \tfrac{1}{2} m r^2 \dot{\phi}^2 - U\right] = m r \dot{\phi}^2 - \frac{\partial U}{\partial r}$$

$$\underbrace{\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{r}} = \frac{d}{dt}\left[m\dot{r}\right] = m\ddot{r}}$$

$$\Rightarrow \quad m r \dot{\phi}^2 - \frac{\partial U}{\partial r} = m\ddot{r}$$

Want to relate to Force, where $\vec{F} = -\vec{\nabla} U = -\dfrac{\partial U}{\partial r}\hat{r} - \dfrac{1}{r}\dfrac{\partial U}{\partial \theta}\hat{\theta}$

$$\Rightarrow \quad F_r = m\ddot{r} - m r \dot{\phi}^2$$

$$\Rightarrow F_r = -\frac{\partial U}{\partial r}, \quad F_\theta = -\frac{1}{r}\frac{\partial U}{\partial \theta}$$

$\Rightarrow$ Newton's 2nd law in polar coordinates.

No need to tediously calculate $\ddot{\vec{r}}$ like in Cal 1 !

ϕ- equation :

$$\frac{\partial \mathcal{L}}{\partial \phi} = -\frac{\partial U}{\partial \phi} = r F_\theta$$

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \frac{d}{dt}\left[m r^2 \dot{\phi} \right]$$

$$\Rightarrow \quad \underbrace{r F_\theta}_{r F_{tangential}} = \frac{d}{dt}\underbrace{\left[m r^2 \dot{\phi} \right]}_{\text{angular momentum } \equiv L}$$

$\Rightarrow$ torque!

$$\Rightarrow \quad \Gamma = \frac{d}{dt} L$$

Multiparticle Systems : With 2 particles, we can write

$$\mathcal{L} = \frac{1}{2} m \dot{\vec{r}}_1^{\,2} + \frac{1}{2} m \dot{\vec{r}}_2^{\,2} - U(\vec{r}_1, \vec{r}_2) = \mathcal{L}(x_1, x_2, y_1, y_2, z_1, z_2, \dot{x}_1, \dot{x}_2, \dot{y}_1, \dot{y}_2, \dot{z}_1, \dot{z}_2)$$

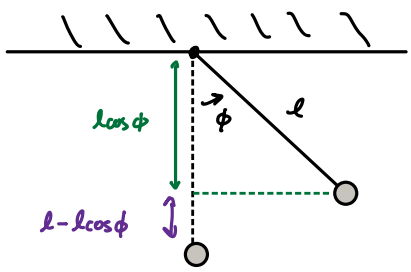
$$\Rightarrow \frac{\partial \mathcal{L}}{\partial x_i} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}_i} \qquad i = 1, 2 \quad \& \; y \quad \& \; z$$

- or -

$$\frac{\partial \mathcal{L}}{\partial q_i} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \qquad i = 1, \dots, 6$$

Constrained Systems

Consider a pendulum :



This is a 1D system that can be completely described by ϕ

$$\mathcal{L} = T - U \qquad T = \frac{1}{2} m \vec{v} \cdot \vec{v} = \frac{1}{2} m (r\dot{\phi})^2 = \frac{1}{2} m l^2 \dot{\phi}^2$$

$$U = mgh = mgl(1 - \cos\phi)$$

$$\mathcal{L} = \frac{1}{2} m l^2 \dot{\phi}^2 - mgl(1 - \cos\phi)$$

$$\Rightarrow \quad \frac{\partial \mathcal{L}}{\partial \phi} = mgl(-\sin\phi) = -mgl\sin\phi$$

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \frac{d}{dt}\left[m l^2 \dot{\phi} \right] = m l^2 \ddot{\phi}$$

$$\Rightarrow \quad \boxed{-mgl \sin\phi = m l^2 \ddot{\phi}} \quad \Rightarrow \quad \left.\begin{aligned} F_g r \sin\phi &= I\alpha \\ \Gamma &= I\alpha \end{aligned}\right\} \text{ reproduces } \Gamma = I\alpha$$

- The only vector analysis we did was finding $\vec{v}$ in polar coordinates. No cross products, etc.

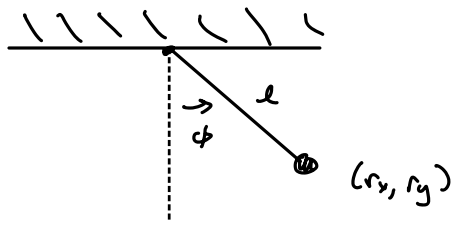
Consider N particles w/ positions $\vec{r}_\alpha$, $\alpha = 1, \dots, N$
Let the system be described by n coordinates $q_1, \dots, q_n$

<u>Generalized coordinates</u>: $\vec{r}_\alpha = \vec{r}_\alpha(q_1, \dots, q_n, t)$ $\quad \alpha = 1, \dots, N$
$q_i = q_i(\vec{r}_1, \dots, \vec{r}_N, t)$ $\quad i = 1, \dots, n$

Assume n is the smallest # that allows this parametrization

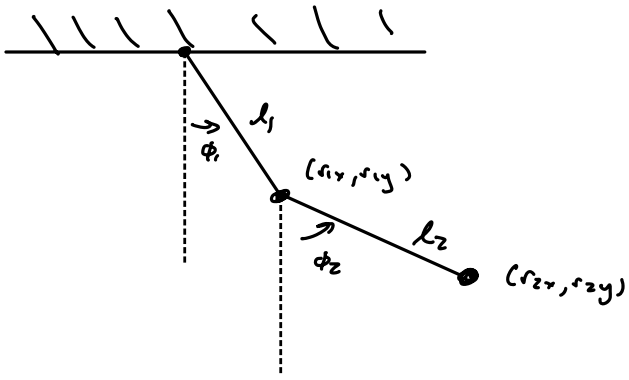
Ex: N gas molecules $\Rightarrow n = 3N$
N atoms forming a rigid body $= n = 6$
$\Big\}$ $n \neq N$ usually

Example:



$r_x = l \sin\phi$
$r_y = l \cos\phi$
$\Rightarrow$ only need ϕ

Example:



$r_{1x} = l_1 \sin\phi_1$
$r_{1y} = l_2 \cos\phi_1$

$r_{2x} = r_{1x} + l_2 \sin\phi_2 = l_1 \sin\phi_1 + l_2 \sin\phi_2$
$r_{2y} = r_{1y} + l_2 \cos\phi_2 = l_1 \cos\phi_1 + l_2 \cos\phi_2$

$\Rightarrow$ the whole system needs only ϕ_1, ϕ_2

<u>Note</u>: Sometimes $r_{\alpha x}$, $r_{\alpha y}$, or $r_{\alpha z}$ depend directly on t (for example, non-inertial reference frames)

Usually they do not.

If there's no explicit time-dependence, then q_i are "<u>natural coordinates</u>"